



KIET Group of Institutions, Delhi-NCR,
Ghaziabad (UP)
Department of Computer Science and Engineering

Face Disease Detection and Analyzer using Deep Learning

Project Synopsis of Major Project

BACHELOR OF TECHNOLOGY
COMPUTER SCIENCE AND ENGINEERING

SUBMITTED BY:

Ankur Kumar (2200290100026)

Abhishek Yadav (220029010008)

Anmol Kumar (2200290100027)

Anshul Rai (2200290100031)

Project Guide: Prof. Pushpendra Kumar Sir

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Abstract

This project aims to develop an automated system for detecting and analyzing facial diseases using deep learning techniques. The system will utilize convolutional neural networks and computer vision algorithms to identify various dermatological conditions from facial images, including but not limited to acne, eczema, psoriasis, rosacea, and skin cancer. The analyzer will provide preliminary diagnoses, severity assessments, and recommendations for further medical consultation. This technology has the potential to improve early detection rates, reduce healthcare costs, and make dermatological screening more accessible.

This project will save time and money of both public and hospital with accurate, fast and efficient results. There are many fields and specialist in medical science, but this one will cover every field and with huge amount of data it's performance will be unmatched.

Introduction

1.1 Introduction -

Facial skin diseases affect millions worldwide, yet many cases go undiagnosed or are identified too late due to limited access to dermatological care. Recent advances in deep learning and computer vision offer promising solutions for automated skin disease detection. This project explores the development of a robust system that can analyze facial images with high accuracy, providing users with immediate feedback about potential skin conditions.

This project focuses on developing a deep learning-based facial disease analyzer that can:

Detect multiple skin conditions from facial images.

Provide severity assessment (mild, moderate, severe).

Suggest preliminary recommendations (e.g., "Consult a dermatologist").

Work on mobile devices for easy accessibility.

1.2 Problem Statement-

- Current challenges in dermatological care include:
- Long wait times for specialist appointments
- High costs of dermatological consultations
- Limited access in rural and underserved areas
- Variability in diagnostic accuracy among practitioners
- Lack of self-assessment tools for early detection
- Lack of early detection tools for common skin diseases.

An automated AI-based system can help bridge this gap by providing fast, low-cost, and accessible preliminary diagnoses.

1.3 Objective-

The primary objectives of this project are: -

- To develop a deep learning model capable of detecting multiple facial skin diseases with high accuracy

- To create a user-friendly interface for image capture and analysis
- To implement severity assessment for detected conditions
- To provide preliminary recommendations based on analysis results
- To compare the performance of different CNN architectures for this specific application

1.4 Scope-

The project will focus on:

- Common facial dermatological conditions (5-10 major diseases)
- Adult patients (18+ years)
- Visible skin conditions (not subdermal)
- Standard lighting conditions for image capture
- Mobile and web-based implementation options

The product will be available in the hospitals where you can use it on consultation with the doctor. Frequent update and modification will be done to enhance the user experience and accuracy of the model.

Literature Review

1. "Dermatologist-Level Classification of Skin Cancer with Deep Neural Networks"

Authors: A. Esteva, B. Kuprel, R. A. Novoa, J. Ko, S. M. Swetter, H. M. Blau, S. Thrun

Publication: *Nature*, 2017

Summary:

This landmark study presents a deep learning system capable of classifying skin cancer with accuracy comparable to dermatologists. The authors trained a convolutional neural network (CNN) based on the Inception v3 architecture on over 129,000 clinical images covering more than 2,000 skin diseases. The system was validated against 21 board-certified dermatologists, and it performed on par in distinguishing between benign and malignant lesions. This work marked a significant step toward AI-assisted dermatological diagnostics, highlighting the potential of CNNs in clinical settings.

2. "Classification of the Clinical Images for Benign and Malignant Cutaneous Tumors Using a Deep Learning Algorithm".

Authors: S. Han, M. Kim, W. Lim, et al.

Publication: *Journal of Investigative Dermatology*, 2018

Summary:

This study proposes a CNN-based deep learning system for classifying skin lesions into multiple disease categories. Using clinical images and metadata, the system was trained and tested on a large dataset and achieved high classification accuracy across 12 classes. The research also emphasized the importance of preprocessing techniques such as color normalization and data augmentation to improve the model's generalization. The authors suggested that integrating deep learning into diagnostic workflows could improve early detection of skin cancers and reduce diagnostic errors.

3. "Skin Cancer Classification Using Ensembles of Deep Convolutional Neural Networks".

Authors: M. Mahbod, G. Schaefer, C. Ecker, A. Ellinger, E. P. Ellinger, A. S. Dehshibi

Publication: *Expert Systems with Applications*, 2020

Summary:

This paper introduces an ensemble learning approach for skin cancer classification using multiple deep CNN models. Pre-trained networks such as ResNet and DenseNet were fine-

tuned on dermoscopic image datasets. The ensemble approach combined predictions from individual models to improve robustness and accuracy. The study demonstrated that ensemble models outperformed single models and highlighted the benefits of combining features extracted from different CNN architectures.

4. "Automated Skin Lesion Diagnosis Using Deep Learning Algorithms"

Authors: M. Tschandl, C. Rinner, P. Apalla, et al.

Publication: *JAMA Dermatology*, 2019

Summary:

This research evaluates the diagnostic accuracy of deep learning models compared to dermatologists in identifying skin lesions. Using the HAM10000 dataset, the authors trained CNNs and assessed their performance in multiclass classification tasks. The study found that the deep learning models achieved diagnostic accuracy similar to or better than many practicing dermatologists. It also discussed the role of model interpretability and highlighted the need for validation on diverse populations before real-world deployment.

5. "Facial Acne Detection and Grading Using Deep Learning"

Authors: Y. Yuan, Z. Yan, Z. Zhang, M. Liu

Publication: *IEEE Access*, 2020

Summary:

This paper presents a deep attention-based convolutional neural network for detecting and grading facial acne severity. The model was trained on annotated facial datasets and used attention mechanisms to focus on lesion-prone areas of the face. The system classified acne into various severity levels and types (blackheads, whiteheads, papules, pustules), achieving high classification accuracy. The authors highlighted the potential of AI in dermatology apps for self-assessment and telemedicine.

Feasibility of the Project

1. Technical Feasibility

- Available Technologies

Deep Learning Frameworks: TensorFlow, PyTorch, Keras (well-supported for medical imaging).

Pretrained Models: ResNet, EfficientNet, DenseNet (proven effective in dermatology AI).

Computer Vision Tools: OpenCV, scikit-image (for preprocessing facial images).

Cloud Deployment: AWS, Google Cloud, and Azure support AI model hosting.

- Data Availability

Public Datasets:

- HAM10000 (10,000+ dermoscopic images).
- DermNet (23,000+ skin disease images).
- ISIC Archive (melanoma and lesion datasets).
- Data Augmentation: Can artificially expand dataset size using rotations, flips, and color adjustments.
- Model Performance Expectations
- Model, Expected Accuracy, Inference Speed, Suitability
- ResNet50, 90-93%, Moderate, High accuracy, good for research
- EfficientNet-B4, 88-91%, Fast, Best for mobile deployment
- DenseNet121, 89-92%, Slow, Good for small datasets

Challenges

- Image Quality Dependency: Requires good lighting and resolution.
- Bias in Datasets: Most datasets lack diversity in skin tones.
- Computational Cost: Training deep learning models requires GPUs.

Conclusion: Technically feasible with existing tools, but requires careful dataset curation and model optimization.

2. Economic Feasibility

Cost Factors

Component, Estimated Cost

- GPU Cloud Training (AWS/GCP), 200–200–500
- Mobile/Web App Development, 1,000–1,000–3,000 (if outsourced)
- Dataset Licensing (if needed), 0–0–500 (most datasets are free)
- Deployment (Server Costs), 20–20–100/month

Potential Revenue Models

Freemium App: Free basic scans, paid detailed reports.

Healthcare Partnerships: Licensing to clinics for preliminary screening.

Research Grants: Funding from medical AI initiatives.

Risks

Regulatory Compliance: May require FDA approval if marketed as a diagnostic tool.

Competition: Existing apps (SkinVision, Ada) already provide similar services.

Conclusion: Economically viable for research purposes; commercialization requires investment and regulatory checks.

3. Legal & Ethical Feasibility

Compliance Considerations

GDPR/Privacy Laws: Must anonymize user data.

Medical Device Regulations:

FDA Class I/II (if used for diagnosis).

CE Marking (for EU deployment).

Ethical Concerns

- Misdiagnosis Risk: Must include disclaimers ("Not a replacement for a doctor").
- Bias Mitigation: Ensure model works across different skin tones.
- Conclusion: Feasible with precautions; legal consultation needed for medical use cases.

4. Operational Feasibility

Development Workflow

Data Collection → Model Training → App Development → Testing → Deployment.

Can be completed in 6-12 months with a small team (2-3 developers).

Challenges

Need for Medical Validation: Dermatologist input required for accuracy testing.

User Adoption: Must ensure ease of use for non-tech-savvy users.

Methodology

In project management, "methodology" refers to the structured approach and principles used to plan, execute, and manage a project. It's like a roadmap that guides project teams from initiation to completion, ensuring they follow a consistent process

1. Data Collection & Preprocessing

Datasets: HAM10000, DermNet, ISIC Archive.

Data Augmentation: Rotation, flipping, brightness adjustments.

Ethical Compliance: Anonymizing patient data, ensuring GDPR compliance.

2. Model Development

CNN Architecture:

- Input Layer: 224x224x3 (RGB images).
- Feature Extraction: Pretrained models (ResNet, EfficientNet).
- Classification Head: Softmax for multi-class prediction.

Training:

- Optimizer: Adam.
- Loss Function: Categorical Cross-Entropy.
- Validation: 80-20 train-test split.

3. System Implementation

Frontend: Flutter/React for mobile/web.

Backend: Flask/Django for API handling.

Cloud Integration: AWS/GCP for model deployment.

4. Evaluation Metrics

- Accuracy, Precision, Recall, F1-Score.
- Confusion Matrix Analysis

Facilities Required

- Hardware: GPU (NVIDIA RTX 3090) for deep learning training.
- Software: Python (TensorFlow, Keras, OpenCV).
- Cloud Services: AWS SageMaker for model deployment.
- Medical Collaboration: Dermatologist validation for accuracy testing.
- Huge bundle of related datasets.

Expected Outcome

- A working AI model with >90% accuracy in disease classification.
- A mobile/web application for user-friendly interaction.
- Research paper/documentation on model performance.
- Potential for commercialization (healthtech startups, clinics).

References: -

1. Esteva, A., Kuprel, B., Novoa, R. A., Ko, J., Swetter, S. M., Blau, H. M., & Thrun, S. (2017). Dermatologist-level classification of skin cancer with deep neural networks. *Nature*, 542(7639), 115-118.

Key Finding: Demonstrated that a CNN could classify skin cancer with accuracy comparable to dermatologists.

2. Han, S. S., Kim, M. S., Lim, W., Park, G. H., Park, I., & Chang, S. E. (2018). Classification of the clinical images for benign and malignant cutaneous tumors using deep learning. *Journal of Investigative Dermatology*, 138(7), 1529-1538.

Key Finding: Evaluated deep learning models for distinguishing benign and malignant skin tumors.

3. Tschandl, P., Codella, N., Akay, B. N., Argenziano, G., Braun, R. P., Cabo, H., ... & Kittler, H. (2019). Comparison of the accuracy of human readers versus machine-learning algorithms for pigmented skin lesion classification: an open, web-based, international, diagnostic study. *The Lancet Oncology*, 20(7), 938-9.

Key Finding: Compared AI performance with human dermatologists in lesion classification.